SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA01**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 **Time: 60 Mins**
No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P Surendra Kumar Reddy

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

INTRODUCTION

1.1 Introduction

Artificial Intelligence (AI) has become an important technology in today's digital world. One of its popular applications is the chatbot, which allows users to communicate with a computer using natural language. Chatbots are widely used in various fields such as education, healthcare, banking, and customer support to provide quick and automated responses.

Educational institutions receive many queries from students regarding courses, admissions, faculty members, fees, and contact details. Answering these questions manually requires time and effort from the administrative staff. Students also expect quick responses without visiting the college office or searching through multiple web pages.

The **Student Support Assistant Chatbot** is developed using **Google Dialogflow ES** to provide instant responses to student queries. It uses **Natural Language Processing (NLP)** to understand the user's questions and respond appropriately. The chatbot is designed to assist students by providing information about **Course Information, Faculty Information, Admission Process, Fee Structure, and Contact Details**.

The chatbot works by identifying the user's intent through predefined training phrases and responding with suitable information. It provides a simple and user-friendly interface where students can interact using normal conversational language. Since the chatbot is available online, students can access information anytime without depending on office working hours.

This project demonstrates how Artificial Intelligence can improve communication between students and educational institutions. It reduces the workload of administrative staff and provides students with accurate information quickly and efficiently.

1.2 Problem Statement

Students often face difficulties in obtaining information related to admissions, courses, faculty, fees, and contact details. They usually have to visit the administration office, make phone calls, or browse the college website, which consumes time and effort. Administrative staff also spend considerable time answering the same questions repeatedly.

To solve this problem, an intelligent chatbot is proposed that can automatically answer student queries using Natural Language Processing. The chatbot provides instant responses, improves communication, and reduces manual workload.

1.3 Objectives

The objectives of the proposed system are:

- To develop a Student Support Chatbot using Google Dialogflow ES.

- To provide instant responses to student queries.
- To reduce the workload of administrative staff.
- To provide information about courses, faculty, admissions, fees, and contact details.
- To improve communication between students and the institution.
- To demonstrate the use of Artificial Intelligence and Natural Language Processing in education.

1.4 Scope of the Project

The Student Support Assistant Chatbot is designed to provide basic information to students through an interactive conversational interface. The chatbot currently includes the following modules:

- Course Information (Java, Python, Artificial Intelligence)
- Faculty Information
- Admission Information
- Fee Information
- Contact Information

The chatbot is easy to use and can be accessed through a web interface. In the future, it can be enhanced by adding voice support, multilingual support, database connectivity, and integration with the college website or mobile application.

SYSTEM ANALYSIS

System analysis is the process of studying the existing system, identifying its limitations, and proposing an improved solution. It helps in understanding the requirements of the project and designing a system that provides better performance, efficiency, and user satisfaction. In this project, the analysis focuses on improving the communication between students and educational institutions by automating responses to frequently asked questions using a chatbot.

2.1 Existing System

In the existing system, students obtain information regarding courses, admissions, faculty details, fee structure, and contact information by visiting the college office, making phone calls, sending emails, or browsing the college website. These methods require manual effort from both students and administrative staff.

Students often experience delays in receiving responses, especially during admission periods when the number of inquiries is high. Searching for information on websites may also be confusing because users have to navigate through multiple pages before finding the required details. In addition, the administration office is available only during working hours, making it difficult for students to receive assistance at any time.

Disadvantages of the Existing System

- Manual response to student queries.
- Time-consuming process.
- Limited availability during office hours.
- Increased workload for administrative staff.
- Delay in providing information.
- Students may not receive immediate assistance.

2.2 Proposed System

The proposed system is a **Student Support Assistant Chatbot** developed using **Google Dialogflow ES**. The chatbot uses **Natural Language Processing (NLP)** to understand user queries and provide appropriate responses automatically.

The chatbot is designed with different intents and training phrases to recognize student questions related to course information, faculty details, admission procedures, fee structure, and contact information. When a user enters a query, Dialogflow identifies the intent and returns the corresponding response. This enables students to receive accurate information instantly without requiring manual assistance.

The chatbot is simple to use, available online, and can assist multiple users simultaneously. It improves communication between students and the institution while reducing the workload of administrative staff.

Advantages of the Proposed System

- Provides instant responses to student queries.
- Available 24×7.
- Reduces administrative workload.
- Easy to use and user-friendly.
- Improves communication between students and the institution.
- Saves time for both students and staff.
- Can be expanded with additional features in the future.

2.3 Feasibility Study

A feasibility study determines whether the proposed system is practical and beneficial.

Technical Feasibility

The chatbot is developed using Google Dialogflow ES, which is a cloud-based platform for creating conversational applications. The required software and internet connectivity are easily available, making the system technically feasible.

Economic Feasibility

The project is cost-effective because Dialogflow provides a free tier suitable for educational projects. No additional hardware or expensive software is required.

Operational Feasibility

The chatbot provides a simple interface where students can interact using natural language. It requires minimal training and can be easily adopted by educational institutions to improve student support services.

2.4 Software Requirements

Software	**Specification**
Operating System	Windows 10/11
Platform	Google Dialogflow ES
Web Browser	Google Chrome
Internet	Required

2.5 Hardware Requirements

Hardware	**Specification**
Processor	Intel Core i3 or above
RAM	4 GB or above
Storage	100 GB free space
Internet Connection	Required

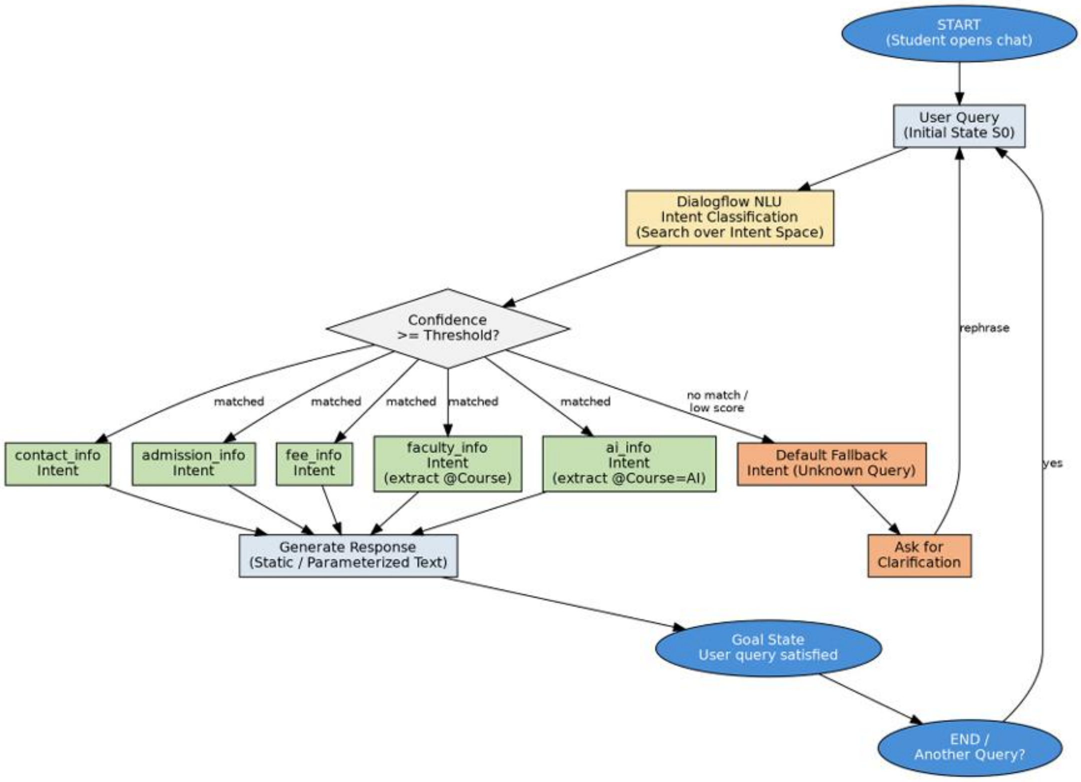
SYSTEM DESIGN

3.1 System Design

System design is the process of defining the architecture, components, and workflow of the proposed system. The Student Support Assistant Chatbot is designed using Google Dialogflow ES to provide quick and accurate responses to student queries. The chatbot uses Natural Language Processing (NLP) to understand user input and identify the appropriate intent before generating a response.

The system consists of different modules such as Course Information, Faculty Information, Admission Information, Fee Information, and Contact Information. Each module is implemented using intents, training phrases, and responses in Dialogflow. When a student enters a query, Dialogflow processes the input, matches it with the appropriate intent, and displays the corresponding response.

The chatbot is designed to be simple, user-friendly, and easily accessible through a web interface. It reduces manual effort by automating the process of answering frequently asked questions.



3.2 System Architecture

Student

User Query/Input

Google Dialogflow ES

Intent Recognition

Response Generation

Student Response

3.3 Modules

The chatbot is divided into five major modules to provide different types of student support services.

Module 1: Course Information

This module provides information about the available courses. In this project, the chatbot responds to queries related to Java, Python, and Artificial Intelligence. Students can ask questions such as "Tell me about Java" or "Explain Python," and the chatbot provides the corresponding course details.

Module 2: Faculty Information

This module provides general information regarding faculty members. Students can ask questions about faculty details or teachers responsible for different subjects. The chatbot responds with predefined faculty information.

Module 3: Admission Information

This module provides information regarding the admission process. It answers questions related to admission procedures, eligibility criteria, required documents, and application details. This helps students understand the admission process without visiting the administration office.

Module 4: Fee Information

This module provides details regarding tuition fees, semester fees, payment methods, and fee structure. Students can quickly obtain fee-related information through the chatbot.

Module 5: Contact Information

This module provides the official contact details of the institution, including office timings, contact number, email address, and administrative office information. Students can use this information to contact the institution whenever necessary.

3.4 Working of the System

The chatbot follows a simple workflow to process user queries.

1. The student enters a question using natural language.
2. Dialogflow receives and processes the user input.
3. The system identifies the appropriate intent using NLP.
4. The chatbot matches the intent with predefined training phrases.
5. The corresponding response is generated.
6. The response is displayed to the student.

This workflow enables the chatbot to provide accurate and quick responses to different student queries.

IMPLEMENTATION

4.1 Implementation

The Student Support Assistant Chatbot was implemented using **Google Dialogflow ES**, a cloud-based conversational AI platform. Dialogflow provides Natural Language Processing (NLP) capabilities that enable the chatbot to understand user queries and respond with appropriate information. The chatbot was developed without complex programming by creating intents, training phrases, and predefined responses.

The implementation process started with creating a Dialogflow agent, followed by configuring intents for different student support services. Each intent was trained using multiple user queries so that the chatbot could recognize different ways of asking the same question. After creating the intents, suitable responses were added and the chatbot was tested using the Dialogflow simulator.

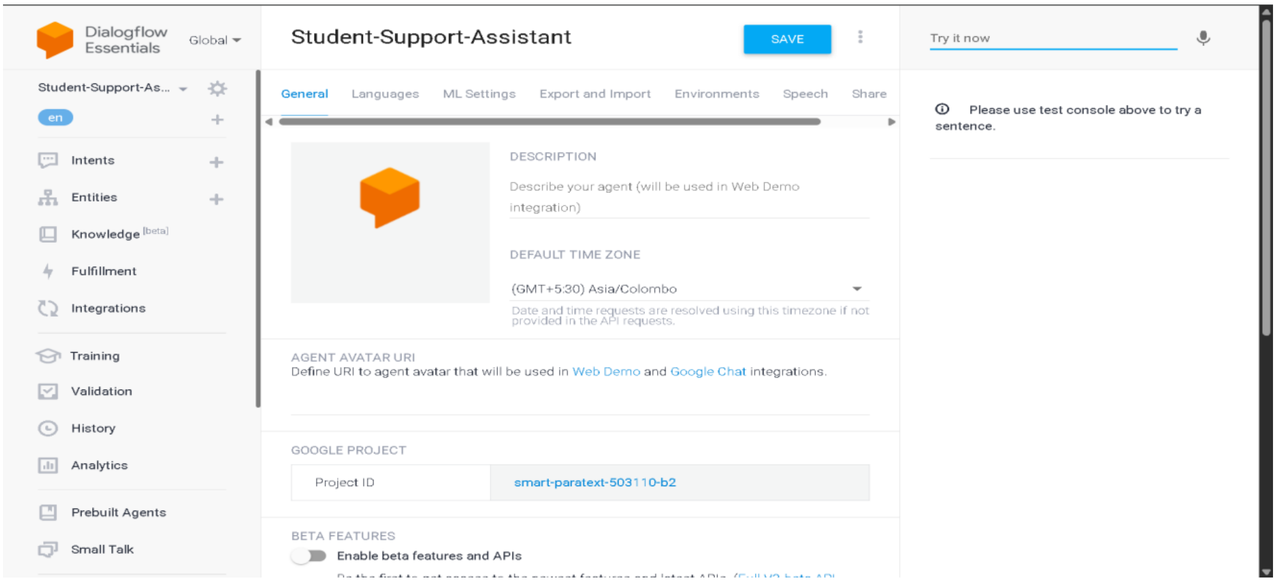
4.2 Tools Used

The following tools and technologies were used to develop the chatbot.

Tool	**Purpose**
Google Dialogflow ES	Chatbot Development
Google Cloud Platform	Hosting Dialogflow Agent
Google Chrome	Accessing Dialogflow Console
Windows 11	Operating System

4.3 Agent Creation

The first step in the implementation process was creating a new Dialogflow ES agent. The agent acts as the main container that stores all chatbot components, including intents, entities, and responses. During agent creation, the project name, language, and time zone were configured.



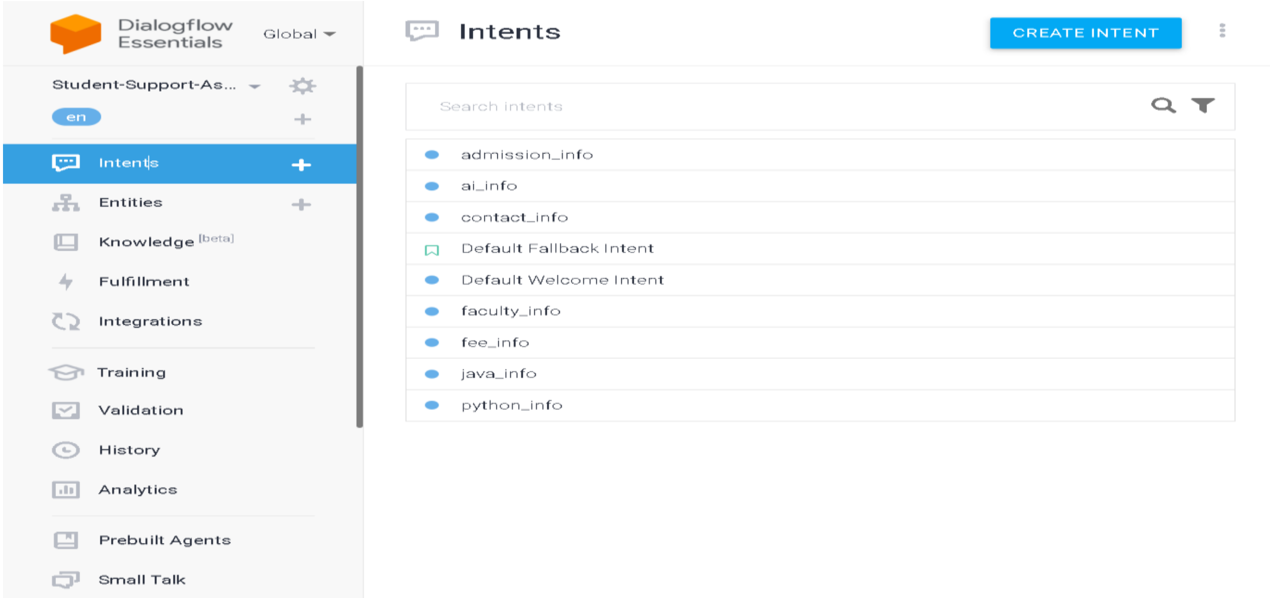
4.4 Intents

An intent represents the purpose of a user's query. Multiple intents were created to handle different student requests.

The following intents were implemented:

- Default Welcome Intent
- Java Information
- Python Information
- Artificial Intelligence Information
- Faculty Information
- Admission Information
- Fee Information
- Contact Information
- Goodbye Intent
- Default Fallback Intent

Each intent contains several training phrases and a predefined response.



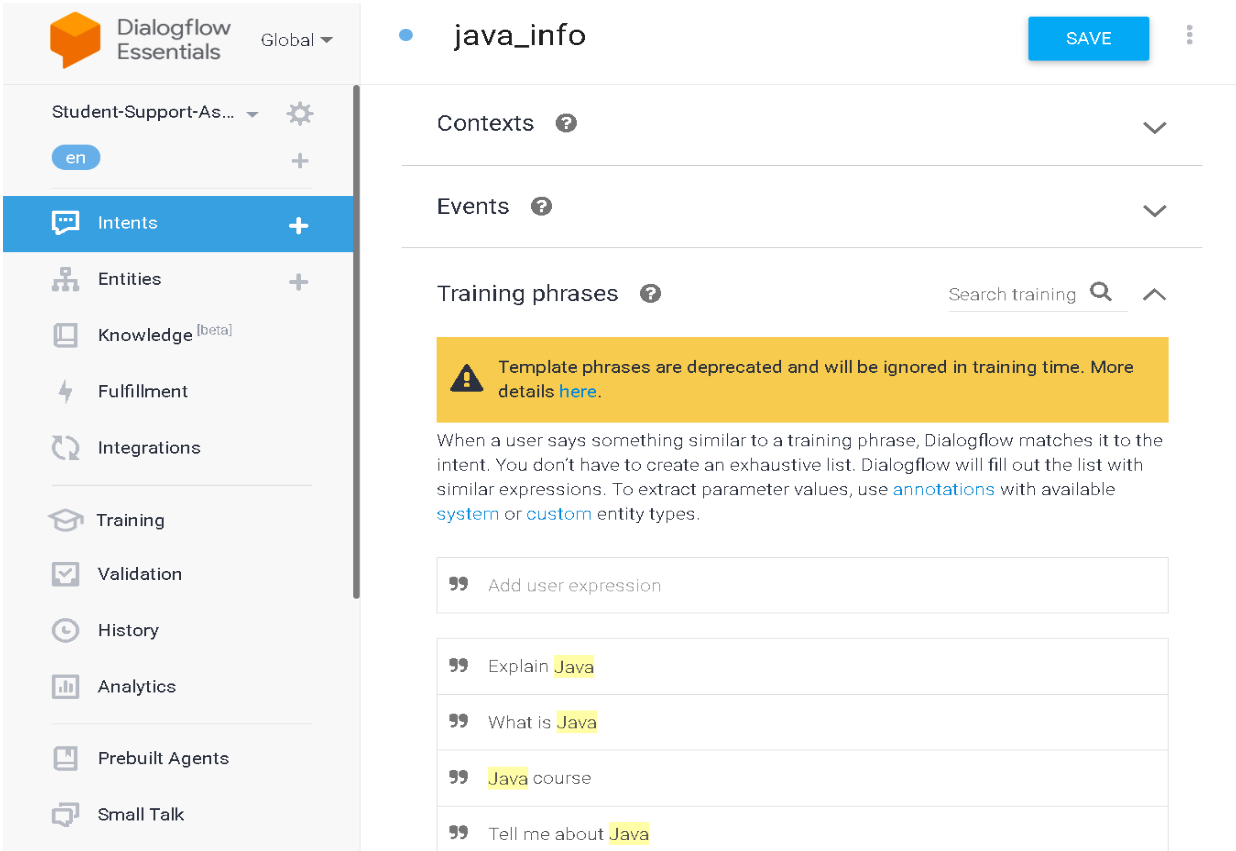
4.5 Training Phrases

Training phrases help Dialogflow understand different ways in which users ask the same question. Multiple training phrases were added for every intent to improve intent recognition accuracy.

Examples:

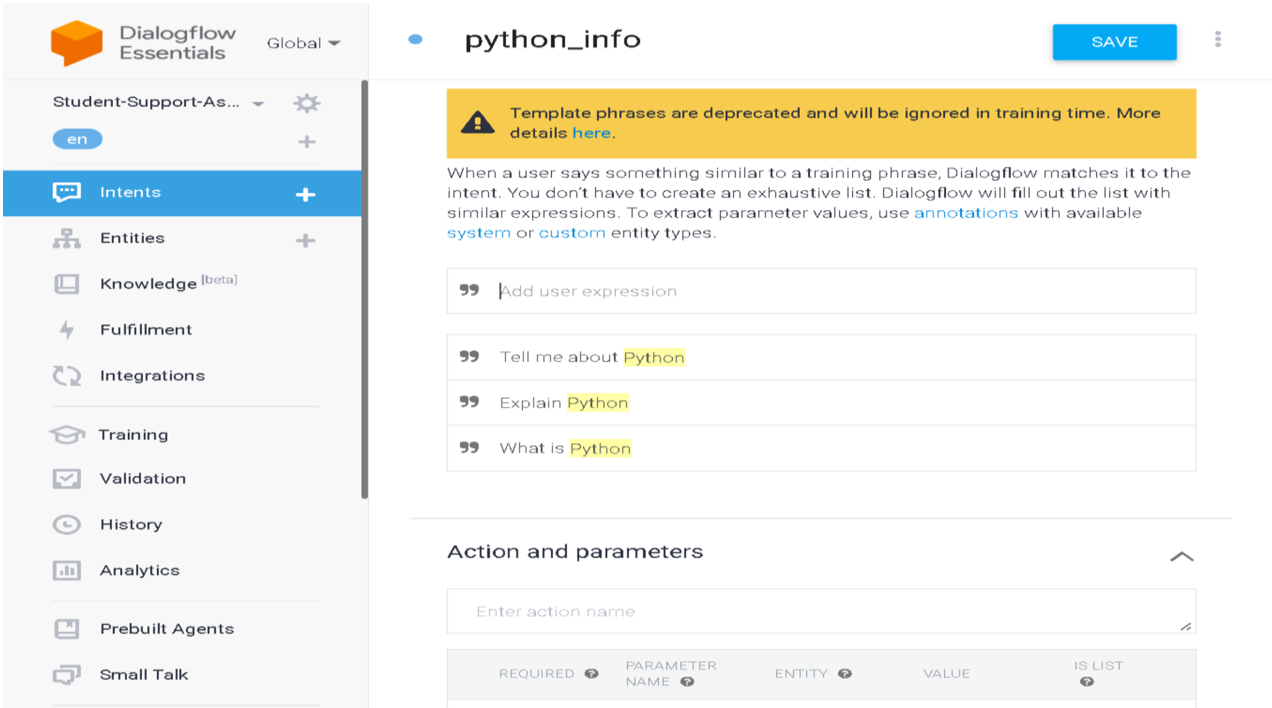
Java Information

- Tell me about Java
- What is Java?
- Explain Java
- Java course



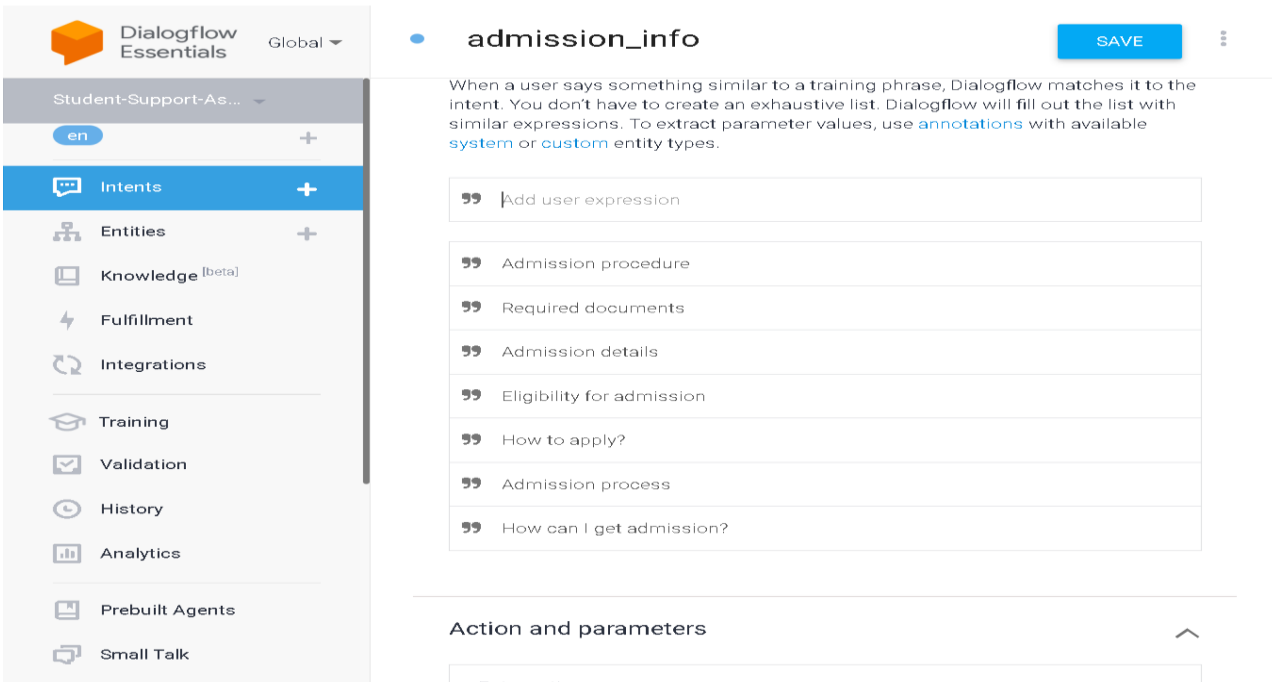
Python Information

- Tell me about Python
- Explain Python
- What is Python?



Admission Information

- Admission process
- How can I get admission?
- Admission details



4.6 Responses

Each intent contains a predefined response that is displayed when the user's query matches the corresponding intent.

Example Responses

Java

Java is an object-oriented programming language used for Android, web, and enterprise applications.

Python

Python is a high-level programming language widely used in Artificial Intelligence, Machine Learning, Data Science, and Web Development.

Admission

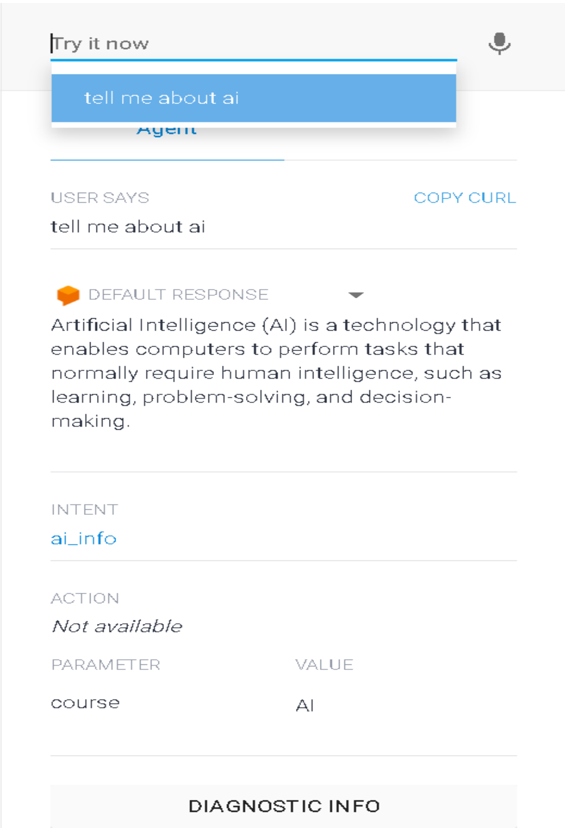
The admission process includes submitting an application form, required documents, and payment of the admission fee.

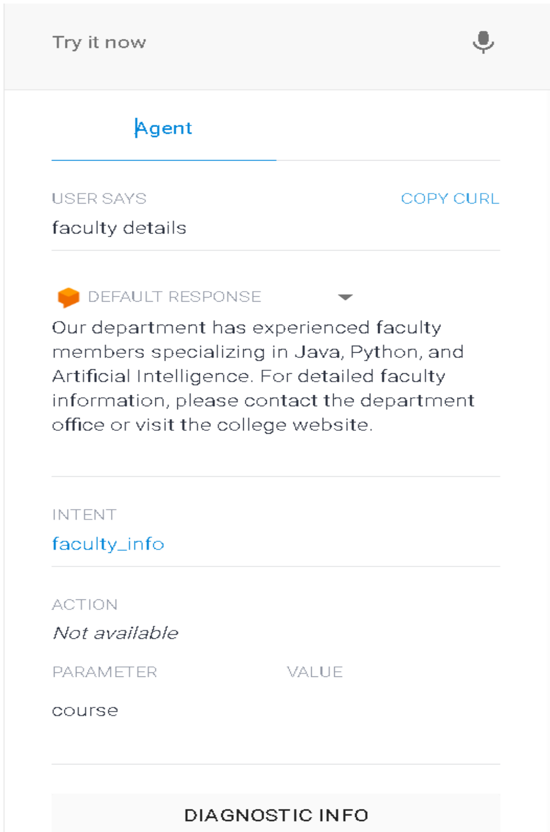
4.7 Testing

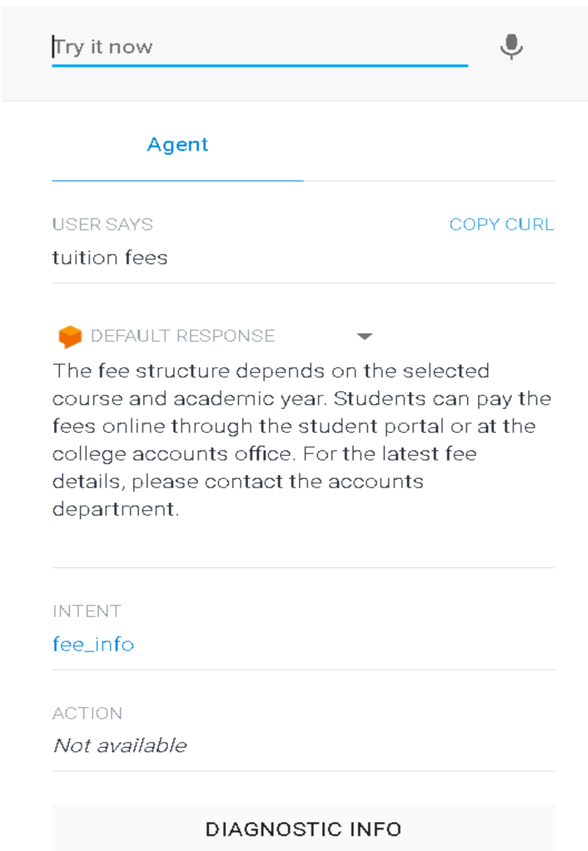
After implementation, the chatbot was tested using the **Dialogflow "Try it now" simulator**. Different user queries were entered to verify whether the chatbot correctly identified the intent and generated the expected response.

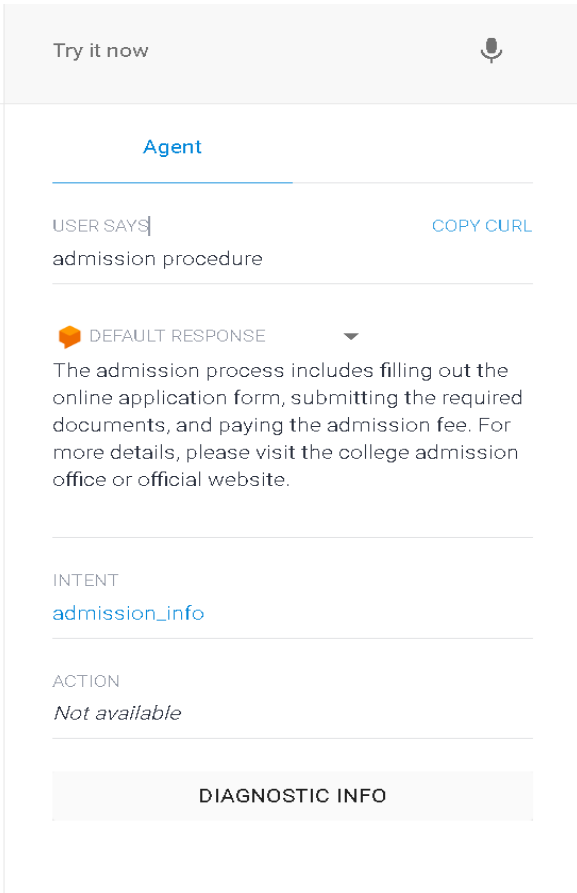
Sample test cases are shown below.

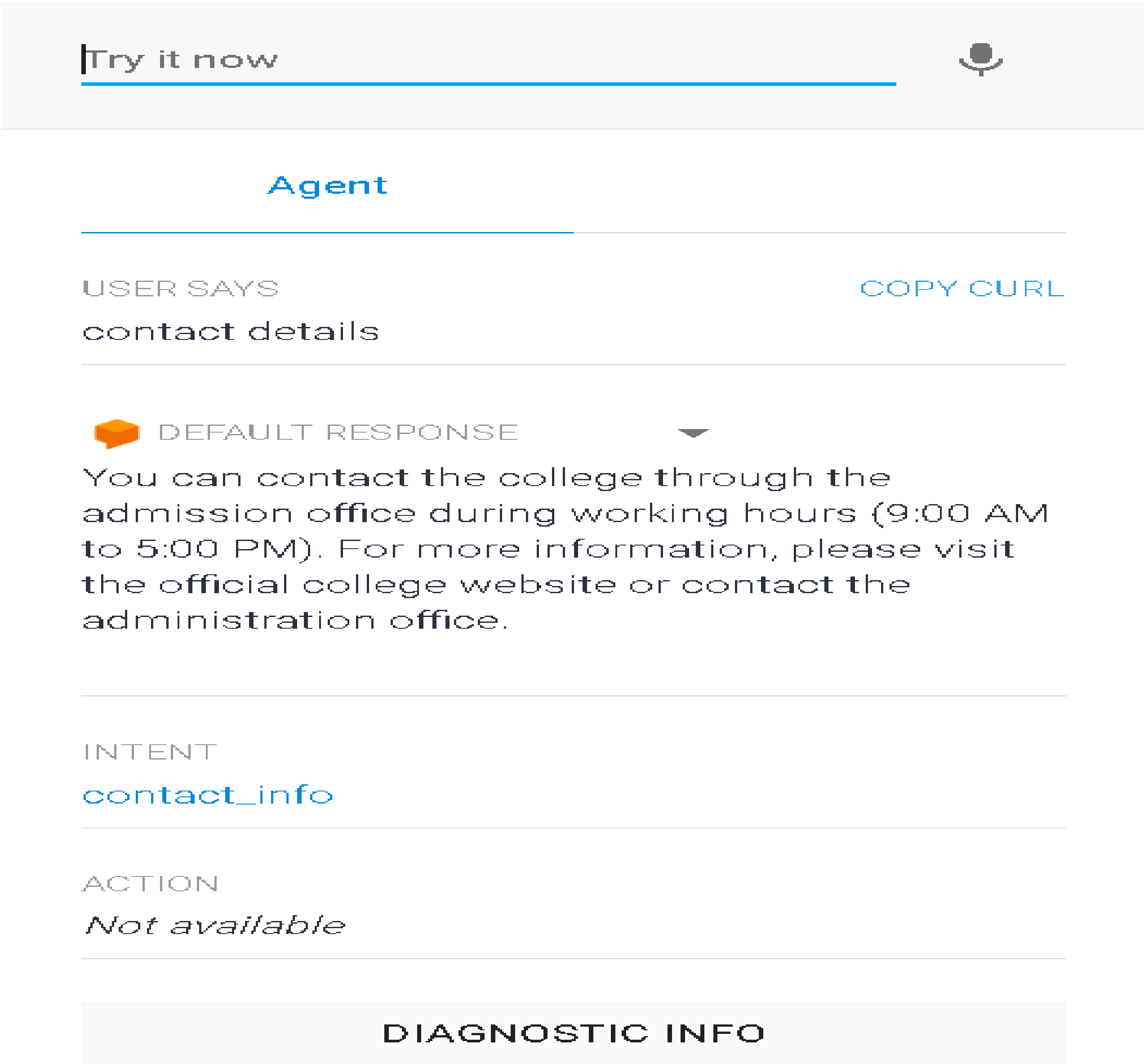
User Query	**Expected Result**	**Status**
Tell me about Java	Displays Java information	Pass
Explain Python	Displays Python information	Pass
Tell me about AI	Displays AI information	Pass
Admission process	Displays admission details	Pass
Fee structure	Displays fee information	Pass
Contact details	Displays contact information	Pass


Try it now
tell me about ai
Agent
USER SAYS
COPY CURL
tell me about ai
DEFAULT RESPONSE
Artificial Intelligence (AI) is a technology that enables computers to perform tasks that normally require human intelligence, such as learning, problem-solving, and decision-making.
INTENT
ai_info
ACTION
Not available
PARAMETER
VALUE
course
AI
DIAGNOSTIC INFO


Try it now
Agent
USER SAYS
COPY CURL
faculty details
DEFAULT RESPONSE
Our department has experienced faculty members specializing in Java, Python, and Artificial Intelligence. For detailed faculty information, please contact the department office or visit the college website.
INTENT
faculty_info
ACTION
Not available
PARAMETER
VALUE
course
DIAGNOSTIC INFO


Try it now
Agent
USER SAYS
COPY CURL
tuition fees
DEFAULT RESPONSE
The fee structure depends on the selected course and academic year. Students can pay the fees online through the student portal or at the college accounts office. For the latest fee details, please contact the accounts department.
INTENT
fee_info
ACTION
Not available
DIAGNOSTIC INFO


Try it now
Agent
USER SAYS
COPY CURL
admission procedure
DEFAULT RESPONSE
The admission process includes filling out the online application form, submitting the required documents, and paying the admission fee. For more details, please visit the college admission office or official website.
INTENT
admission_info
ACTION
Not available
DIAGNOSTIC INFO



4.8 Implementation Summary

The Student Support Assistant Chatbot was successfully implemented using Google Dialogflow ES. The chatbot accurately recognized student queries using predefined intents and training phrases. Testing confirmed that the chatbot responded correctly for all implemented modules, including Course Information, Faculty Information, Admission Information, Fee Information, and Contact Information. The implementation demonstrates the effectiveness of Dialogflow in building intelligent conversational systems for educational institutions.

Conclusion

The **Student Support Assistant** developed using **Google Dialogflow** demonstrates how Artificial Intelligence can effectively automate common university support services. By applying Natural Language Understanding (NLU), intent recognition, and entity extraction, the chatbot provides accurate and timely responses to student queries related to admissions, fees, faculty information, and course details. The implementation of a BFS-style search strategy ensures efficient intent selection, while the fallback mechanism handles unknown queries gracefully.

The evaluation results indicate that the chatbot performs reliably for predefined scenarios, offering quick response times and improved user experience. This intelligent assistant reduces the workload of administrative staff, ensures 24×7 availability, and enhances communication between students and the institution. In the future, the system can be further improved by integrating real-time university databases, multilingual support, advanced contextual conversations, and analytics for continuous learning. Overall, the proposed solution is an effective, scalable, and practical AI-based student support system that meets the growing needs of modern educational institutions.